First Report of *Fusarium boothii* and *Fusarium sudanense* Causing Ear Rot of Maize in Mexico

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Maize ear rot caused by *Fusarium* species reduces yield and compromises grain quality through mycotoxin contamination. During the 2023 and 2024 growing seasons, symptomatic maize ears with whitish to reddish fungal growth and partial kernel decay were collected from 14 representative commercial fields in the Bajío region (Guanajuato, Jalisco, and Michoacán, Mexico), which is a major maize-producing area with a history of ear rot disease. Ten to fifteen ears per field were randomly sampled, manually shelled, and pooled to create a

composite sample. From each sample, 48 kernels (three replicates), without prior classification based on symptom expression, were surface disinfested in 1% sodium hypochlorite for 2 min, rinsed twice in sterile distilled water, and incubated in moist chambers at 25°C for 5 to 7 days. Colonies with *Fusarium*-like morphology that developed from the kernels were subcultured onto potato dextrose agar and purified by single-spore isolation (Leslie and Summerell 2006). Two morphologically distinct *Fusarium* colony types were consistently observed. One produced abundant white aerial mycelium that developed reddish to burgundy pigmentation; its macroconidia were hyaline, falcate, with four to seven septa, measuring 32.5–65.1 × 2–4 µm, and microconidia and chlamydospores were absent. The other produced white to pinkish aerial mycelium; its macroconidia were falcate with three to five septa, measuring 22–53.5 × 2.4–4.3 µm, with abundant unicellular microconidia in short chains and intercalary chlamydospores. Genomic DNA from representative isolates was used to amplify partial sequences of the translation elongation factor 1-α (TEF1-α) and RNA polymerase II subunit (RPB2) (O'Donnell et al. 2022). Sequences were assembled, aligned with reference datasets, and analyzed using maximum-likelihood phylogeny of concatenated TEF1-α and RPB2 under the GTR+G model with 1,000 bootstrap replicates using MEGA11 (Tamura et al. 2021). Phylogenetic analyses confirmed the morphological identification, showing that the first colony type corresponded to *Fusarium boothii* (a member of the *Fusarium graminearum* species complex, FGSC) and the second to *Fusarium sudanense* (a member of the *Fusarium fujikuroi* species complex, FFSC). Representative sequences were deposited in GenBank under accession numbers PX894867, PX894887, PX894860, and PX894880. Pathogenicity was evaluated under field conditions in 2025. A conidial suspension (5×10^5 conidia ml^{-1}) from 7-day-old cultures was injected into maize ears 7 days after silking, while control ears received sterile distilled water. For each species, 10 ears per

replicate were inoculated with three replicates. Symptoms appeared within 21 days, resembling those observed in naturally infected ears. The fungi were re-isolated from symptomatic kernels, and control ears remained asymptomatic. Morphological and molecular characteristics of the reisolated isolates matched the original cultures, fulfilling Koch's postulates. Although *F. boothii* has been reported in Mexico as a causal agent of *Fusarium* head blight of wheat (Malihipour et al. 2012) and as a cause of maize ear rot in other regions of the world (Belizán et al. 2019), *F. sudanense* has been reported as a causal agent of seed rot and seedling blight in wheat (Larran et al. 2020). This is the first report of *F. boothii* and *F. sudanense* causing maize ear rot in Mexico.

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